

MedTech ERP: Integrated Quality Crisis Management

End-to-End Workflow Scenario:
Cardiac Pacemaker CP-2000



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Cardiac Pacemaker CP-2000



THE MISSION:

How 7 integrated modules managed a critical quality event—from discovery to resolution—in exactly 16 weeks.

CONTEXT:

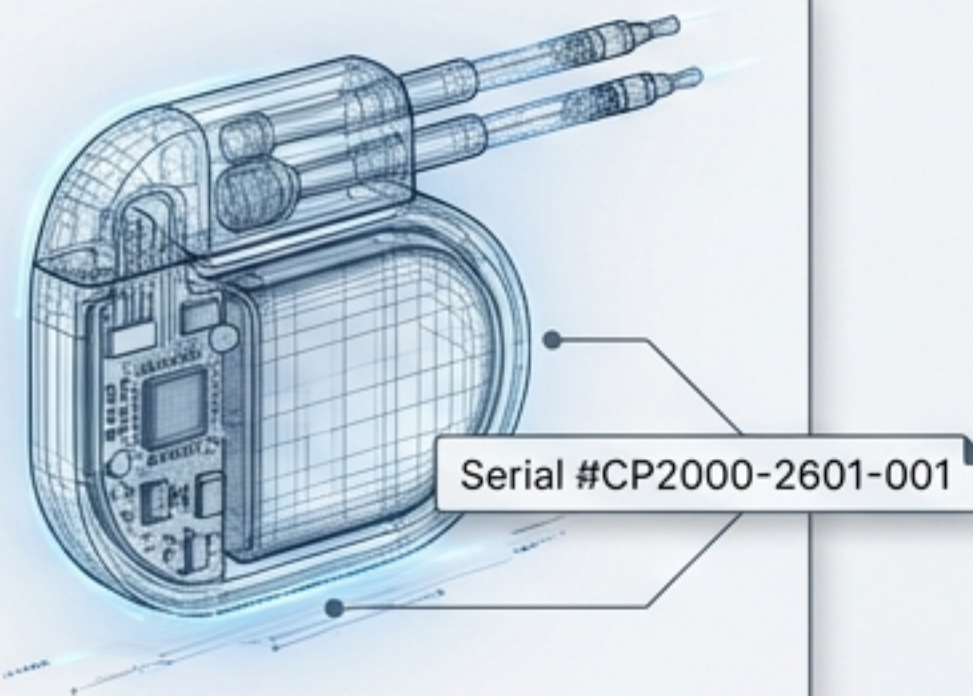
Based on a real-world compliance workflow.

The Scenario: Preventing Patient Harm Through Digital Traceability



Spec Sheet

Device:	Cardiac Pacemaker CP-2000 (Risk Class III)
The Crisis:	Intermittent battery connection failure in Lot LOT-2601-A
The Outcome:	100% Recovery. 0 Patient Harm.



Jan 10: Manufacturing

Jan 15: Defect Discovery

Jan 25: Recall

Apr 30: Resolution

Week 1: Pre-Production

Phase 1: The Compliance Foundation

****Vendor Validation**:**
Purchase Order PO-2025-1234 automatically approved only after system validates BatteryTech's ISO 13485 status.

BatteryTech Solutions Inc.

Vendor/Supplier: BatteryTech Solutions Inc.

Certification Type

Certificate Number

NOTES

PRODUCT CATEGORIES COVERED

CERTIFICATE

Certificate Document

DEVICE CLASSIFICATION

Is Medical Device ☒

Device Family

Risk Classification: Class III

****Regulatory Flags**:**
Product CP-2000 defined as Class III, unlocking strict traceability requirements.

Jan 10: Manufacturing Execution

Jan 15: Defect Discovery

Jan 25: Recall

Apr 30: Resolution

Week 2: Production Execution

Manufacturing the Digital Twin

DHR: Cardiac Pacemaker CP-2000 / CP2000-2601-001

DEVICE IDENTITY

Serial/Lot Number CP2000-2601-001

Device Master Cardiac Pacemaker CP-2000

UDI Record

MANUFACTURING

Manufacturing Order

Manufacturing Date ¹

Components & Genealogy

Distribution

DHR Compilation

Component

Component Lot/Serial

Battery BT-9900

BT-9900-LOT456

Battery BT-9900

BT-9900-LOT456

(01)00312345678906

UDI COMPONENTS

UDI-PI (Production Identifier) ²

Serial Number

Lot/Serial ³

SH:CP2000-2601-001 MFG:20260110 EXP:20260110

CP2000-2601-001

PRODUCT INFO

Product Variant

Manufacturing Date

Expiry Date

****Granular Traceability****: The exact battery lot (BT-9900-LOT456) is permanently bonded to this specific pacemaker's digital record.

Jan 10: Manufacturing

Jan 12-20: Distribution

Jan 15: Defect Discovery

Apr 30: Resolution

Distribution: Maintaining Chain of Custody



UDI Record

Components & Genealogy Distribution BHR Compilation

Final Customer

Ship Date

Final Customer
Memorial Hospital

CP2000-2601-001

CP200G-2601-002

CP2000-2601-001



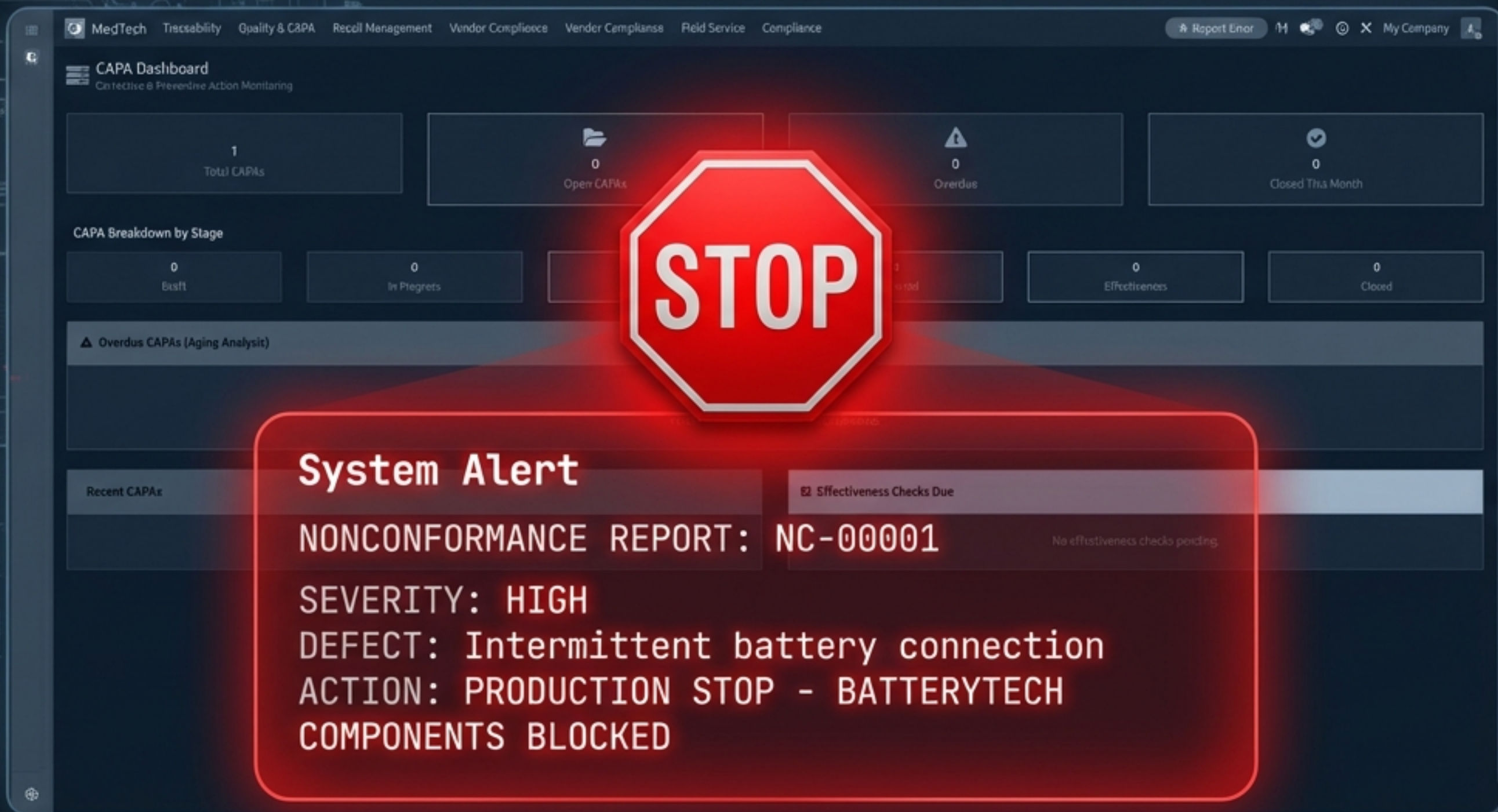
The system **retains a live link between** the DHR and the **specific customer location**, turning 'sold' inventory into 'trackable' assets.

Jan 13: Quality Issue Is Discovery

Jan 15: Quality Issue Discovery

Jan 16: Quality Issue Control

The Alarm: Nonconformance Detected



Field Intelligence: Verifying the Risk



Serial ID	Device ID	Customer	Device Type	Qty Due	Location	Status
93 0201	CP2000-2601-001	Memorial Hospital	Pre-implant Diagonal	1x IL 500	Lot C: 000	Captured Outbreak

****TECHNICIAN REPORT - SV-000001****

Location: Memorial Hospital

Unit: CP2000-2601-001 (Pre-implant)

Finding: Resistance 2.5Ω (Spec <0.5Ω)

Action: DEVICE QUARANTINED ON-SITE

Real-time field data confirms the manufacturing defect exists in shipped units.

Jan 25: Crisis Response

The Recall: Executing the Containment

COMMAND CENTER

MedTech Traceability Quality & CAPA Recall Compliance Vendor Compliance Field Service Compliance

Recall Products Search...

REC-00001

Recall Process

REC-00001

Recall Details Recall Overview Recall Status **BLOCKED** Status Pending Recall Process Action

Recall Process

Recall Process

12 Customer Notifications Sent | 3 Warehouse Units Auto-Blocked

Continue Details

SCOPE: Lot LOT-2601-A (15 Units)

ACTION: 12 Customer Notifications Sent
| 3 Warehouse Units Auto-Blocked

WAREHOUSE LOCKDOWN

MedTech Traceability Quality & CAPA Vendor Compliance Field Service Compliance

Quarantines Search...

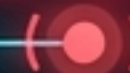
Lot/Serial	Product	Quantity	Quarantine Location	Status
Lot-2601-A	CP2008 Device	3 Units	WH-LOCKDOWN	BLOCKED
Lot-3601-A	CP2008 Device	3 Units	WH-LOCKDOWN	BLOCKED
Lot-0601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-4601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-5601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-6601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-7601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-8601-A		3 Units	WH-LOCKDOWN	BLOCKED
Lot-9601-A		3 Units	WH-LOCKDOWN	BLOCKED

Jan 15: Defect Discovery

Jan 22: Service Visit

Jan 25: Crisis Response

Feb 01: Root Cause Analysis



Root Cause Analysis: The '5 Whys'

Problem: Battery connection failure

Why? Weak solder joints

Why? Inconsistent terminal plating

Why? Supplier changed plating vendor without notice

ROOT CAUSE: Inadequate supplier
change control process

CONFIDENTIAL

The system moves the team from symptom (solder failure)
to systemic cause (supplier agreement gap).

Jan 15: Defect Discovery

Jan 22: Service Visit

Jan 25: Crisis Response

Feb 01: Root Cause Analysis

Mar 01: Remediation

Remediation: Closing the Loop

Quality Agreement
Rev B

UPDATED

Requires Change
Notification

Suspended

Audited

Reinstated

****Corrective Action:** Enhanced Incoming Inspection (IQP-005) and updated Quality Agreement implemented before vendor reinstatement.

CONFIDENTIAL

Jan 15: Defect Discovery

Jan 22: Service Visit

Jan 25: Crisis Response

Feb 01: Root Cause Analysis

Apr 15: Verification

Effectiveness Check: Validating the Fix

13.3%
Defect Rate

0.0%
Defect Rate

Lot 2601

Validation Lots (150 Units)

CAPA CLOSED

****Recall Effectiveness:: 100% (15/15 Units Recovered)**
Patient Safety:: 0 Patient Harm

CONFIDENTIAL

Jan 15: Defect
Discovery

Jan 22: Service
Visit

Jan 25: Crisis
Response

Feb 01: Root
Cause Analysis

Apr 15:
Verification

Compliance
Audit

The Compliance Backbone: Audit-Ready by Default

User: Quality Director | Action: Approve Recall | Timestamp: Jan 25, 14:00 UTC

User: System | Action: Log Entry Created | Timestamp: Feb 25, 09:00 PM

User: OdooBot | Action: Record created | Timestamp: Feb 25, 09:00 PM

User: Admin | Action: User Login | Timestamp: Feb 25, 08:45 PM

User: System | Action: Log Entry | Timestamp: Feb 25, 14:00 UTC

User: Admin | Action: User Login | Timestamp: Feb 25, 08:45 PM

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****21 CFR Part 11 Compliance:****
Every status change, approval, and record edit is permanently logged with 'Who, What, When' traceability.

Jan 15: Defect
Discovery

Jan 22: Service
Visit

Jan 25: Crisis
Response

Feb 01: Root
Cause Analysis

Apr 15:
Verification

System
Architecture

System Integration Architecture



Zero data re-entry. One single source of truth
from factory floor to field recovery.

Key Learnings & ROI



Speed:
Lot segregation in
hours, not days.



Integration:
Field data directly
directly triggered
Quality Actions.



Safety:
Zero patient
exposure achieved.



Compliance:
100% audit-ready
documentation.

Crisis Resolved. Compliance Assured.

MedTech ERP



"The system successfully managed a complex quality crisis while protecting patient safety."

Contact us for full system specifications.